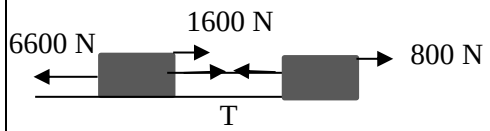
Question			Answer	Marks	AO	Guidance
9	(a)		Horizontal forces (could be mirror image)  	**B1** **B1** **[2]**	**3.3** **3.3**	Ignore vertical forces. 6600 N and resistances correctly placed and labelled. No extra horizontal forces Common tension in the towbar shown.
9	(b)		Tractor and trailer together $6600-1600-800=2800a$ $a=1.5\ \text{ms}^{-2}$ Newton's second law for the trailer $T-800=1000\times1.5$ $T=2300$ N	**M1** **A1** **M1** **A1**	**3.1b** **1.1b** **3.1b** **1.1b**	2800 kg used. Allow a missing or an extra horizontal force for the method mark soi Newton's second law with correct mass. FT their acceleration Also allow for the tractor $6600-T-1600=1800\times1.5$ used cao
			Alternative method Newton's second law for the tractor $6600-T-1600=1800a$ Newton's second law for the trailer $T-800=1000a$ Simultaneous equations giving T = 2300 N	 **M1** **A1** **M1** **A1** **[4]**	 	 Newton's second law for both parts of the system with correct masses; allow a missing or an extra horizontal force for the method mark Both equations correct Attempt to solve simultaneous equations leading to a value of T BC means method need not be seen cao